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Think Like a Well, Act Like a Facility: The Autonomous Fusion of Subsurface and Surface

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Abstract

This paper presents a field-driven enhancement to ROBOWELL[®], ADNOC Offshore's Autonomous Well Control system, designed to optimize oil and gas production operations. The objective of this study is to address a recurring operational challenge: optimizing well performance without exceeding the handling capacity of surface facilities, particularly the slug catcher. During ROBOWELL[®] deployment, the system demonstrated effective well optimization; however, instances were observed where increased gas production exceeded slug catcher limits, leading to surface instability. To mitigate this, a control logic module was developed to incorporate real-time slug catcher level data into the autonomous decision framework. This logic continuously monitors downstream buffer capacity and dynamically constrains well adjustments based on available processing headroom.

The new module was integrated into the existing Advanced Process Control (APC) framework, enabling closed-loop coordination between reservoir inflow and surface processing readiness. As a result, the system now modulates its optimization behavior in response to downstream conditions, holding back when slug catcher levels are high and resuming optimization when capacity is available.

Field results demonstrated a significant reduction in operational upsets and manual interventions, with sustained production optimization that respected downstream thresholds. This advancement redefines the role of autonomous well control from a localized optimizer to a facility-aware orchestrator, marking a step change in digital oilfield automation. The integration of upstream and downstream logic within a single autonomous system represents a novel approach in field operations. It validates the potential for control systems to make facility-aware decisions without human input, paving the way for fully connected, autonomous production environments that maximize value across the hydrocarbon value chain.

This work contributes to the broader vision of intelligent field automation by demonstrating how real-time data integration across production domains can enhance system responsiveness and stability. The approach offers a scalable model for other facilities seeking to align subsurface optimization with surface constraints and sets a precedent for future developments in autonomous oilfield control systems.

Introduction

In the evolving landscape of upstream automation, ADNOC Offshore has been at the forefront of deploying autonomous technologies to enhance operational efficiency and production optimization. ROBOWELL[®], an autonomous well control system, has demonstrated significant success in optimizing well performance through real-time, model-based decision-making. The foundational work presented by [Hernandez et al. \(2024\)](#) introduced the implementation of Multivariable Predictive Control (MPC) within ROBOWELL[®], showcasing its ability to manage complex well dynamics and enforce operational constraints across natural flow and gas lift wells.

However, a recent field challenge revealed a critical limitation in conventional upstream logic: the absence of downstream awareness in the autonomous control system. During deployment, ROBOWELL[®] consistently optimized gas lift wells based on reservoir and surface parameters. Yet, in specific scenarios, it inadvertently pushed more produced gas than the slug catcher could accommodate, leading to instability and operational upsets. Rather than introducing manual overrides or additional supervisory layers, the team pursued a more integrated solution—enhancing the system's logic to incorporate real-time slug catcher level data.

This enhancement marked a pivotal shift in automation philosophy. By embedding downstream constraints directly into the Advanced Process Control (APC) framework, ROBOWELL[®] evolved from a well optimizer into a value chain orchestrator. The system now continuously monitors slug catcher levels and dynamically adjusts well outputs only when downstream capacity permits. This closed-loop, hands-off logic ensures that decisions are not only optimal for individual wells but also harmonized with facility readiness.

Field results demonstrated a substantial reduction in process upsets, improved system stability, and a notable decrease in manual interventions. More importantly, the integration fostered greater confidence among field teams, as the system began making decisions that reflected a holistic understanding of both subsurface and surface dynamics.

This paper presents the methodology, implementation, and performance outcomes of this breakthrough enhancement. It showcases how bridging upstream and downstream intelligence within a single autonomous framework can redefine the role of digital systems in oilfield operations, paving the way for fully connected, facility-aware automation that delivers maximum value from reservoir to export.

Statement of Theory and Definitions

Theory of Constraint-Aware Autonomous Control

The evolution of autonomous well optimization systems is grounded in the principles of Advanced Process Control (APC), with Multivariable Predictive Control (MPC) serving as a core enabler. ADNOC Offshore's automation strategy leverages MPC to anticipate system behavior, optimize control actions, and enforce operational constraints. Traditionally, these systems have focused on subsurface and surface parameters in isolation, overlooking the dynamic limitations of downstream facilities.

This siloed approach can lead to operational disconnects, particularly in high-throughput environments, where surface processing units such as slug catchers impose critical capacity constraints. Without real-time facility feedback, autonomous systems may inadvertently trigger process upsets, pressure fluctuations, or alarm conditions.

To overcome this, a constraint-aware control logic was implemented that extends the MPC framework to include downstream facility state feedback. Specifically, slug catcher liquid level measurements were integrated as governing constraints within the ROBOWELL[®] optimization algorithm. This enables dynamic coordination between well output and facility capacity, reducing operational risk and enhancing system stability.

Key Definitions

- **Slug Catcher:** A vessel or pipeline segment installed at the surface to receive, separate, and temporarily store intermittent surges of liquid and gas (slugs) arriving from upstream wells, with the primary function of reducing flow fluctuations and preventing downstream process upsets.
- **Choke Movement:** The adjustment of a well's surface choke valve to regulate the produced fluid flow rate, typically as part of closed-loop well optimization strategies.
- **GOR (Gas-Oil Ratio):** The volumetric ratio of produced gas to oil, expressed as standard cubic feet per barrel, serving as a key indicator of well performance and a determinant of flow regime and facility handling requirements.
- **Freeze Logic:** A control strategy in which further choke adjustments are suspended when the monitored slug catcher level exceeds pre-defined operational thresholds, preventing system overload and protecting processing facilities.
- **Dynamic Oil Target Window:** An adaptive range about the oil production target that allows for transient deviations, facilitating smoother choke modulation and minimizing frequent control actions under fluctuating process conditions.

Application and Relevance

The theoretical basis for this integrated control approach can be traced to the broader literature of process systems engineering and real-time optimization, notably in the context of chemical and petrochemical process industries. However, its application in oilfield digitalization and smart field automation is an emerging domain. Early deployments, such as the constraint-aware control logic applied to ROBOWELL® — highlight the importance of real-time, facility-aware feedback in achieving end-to-end field stability.

The Field Challenge that Sparked a Shift

During routine deployment of ROBOWELL® across ADNOC Offshore assets, a recurring operational anomaly was observed. While ROBOWELL® was effectively optimizing well performance based on reservoir and surface parameters, it occasionally triggered downstream instability by pushing more produced gas than the slug catcher could accommodate. These uncoordinated choke movements, executed without visibility into slug catcher status, led to elevated levels, alarm conditions, and manual interventions that disrupted the otherwise autonomous workflow.

This challenge highlighted a critical gap in the system's logic: the absence of downstream awareness. Instead of introducing additional manual checks or conservative constraints, the team posed a transformative question; what if ROBOWELL® could "listen" to the slug catcher? This question catalyzed a collaborative effort between ADNOC Offshore, AIQ, and APC vendors to redesign ROBOWELL® control logic. The solution was implemented in two key phases:

- **Slug Catcher Level Criteria Integration:** The system was enhanced to monitor real-time slug catcher levels and freeze choke movements when predefined thresholds were exceeded. This logic was applied uniformly across wells with both high and low GOR profiles, ensuring consistent behavior across the field.
- **Dynamic Oil Target Window:** A variable oil target range was introduced to smooth choke transitions, enabling adjustments to choke movements based on fractional oil deficits. Additionally, it halts adjustments once the target falls within an acceptable range. This prevents abrupt fluctuations and maintains system stability.

The result was a closed-loop, facility-aware control system that harmonizes upstream optimization with downstream readiness. By embedding slug catcher constraints directly into the APC framework,

ROBOWELL® evolved from a well-centric optimizer into a value chain orchestrator, capable of making autonomous decisions that reflect the whole production system.

This shift not only resolved the immediate operational issue but also redefined the role of autonomous control in oilfield operations. It demonstrated that upstream systems can be made context-aware, bridging the gap between reservoir performance and surface facility constraints to deliver smarter, safer, and more stable production outcomes.

Building the Bridge: Integrating Surface Awareness

The integration of surface facility constraints, specifically slug catcher level monitoring, into ROBOWELL® existing Advanced Process Control (APC) framework is achieved through a two-step control enhancement aimed at preventing surface processing disturbances caused by uncoordinated choke adjustments.

Slug Catcher Level Freeze Logic

A key control intervention involves embedding a freezing logic module within the system that actively monitors slug catcher liquid levels in real time. When these levels exceed predefined safety or operational thresholds set in consultation with ADNOC operations, the system automatically suspends further choke valve movements across both high and low Gas-Oil Ratio (GOR) wells. This targeted freeze mechanism prevents the exacerbation of slug catcher surges and minimizes pressure disturbances downstream, ensuring the facility operates within its designed handling capacity (see Figure 1).

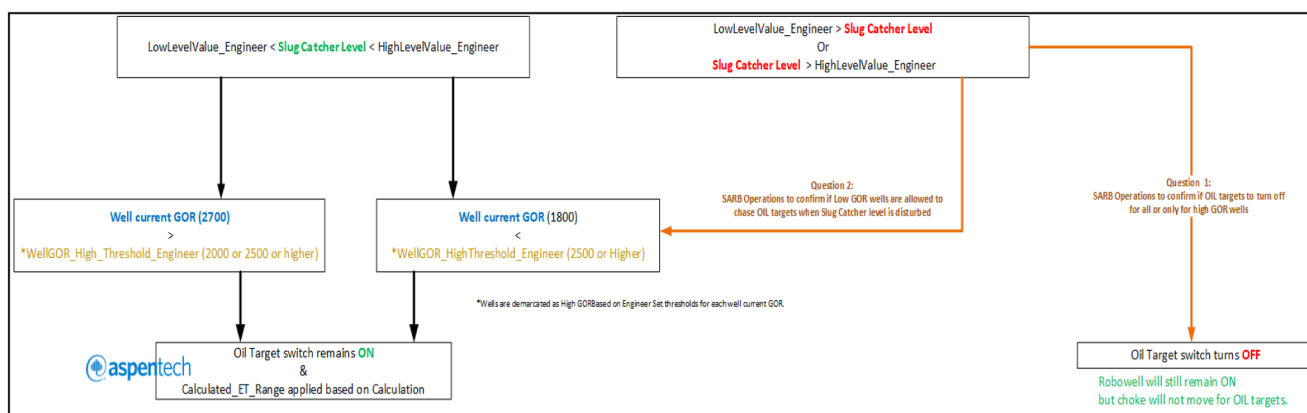


Figure 1—Slug catcher level criteria consideration in ROBOWELL®

This freezing logic is implemented as an overriding constraint in the MPC-based choke position optimizer. It functions as a hard limit condition: once triggered, all choke adjustments are temporarily halted until slug catcher levels normalize below the threshold, thereby aligning well optimization objectives with facility integrity.

Dynamic Oil Target Window (Variable Range Based Control)

Complementing the freezing logic, ROBOWELL® incorporates a dynamic oil production targeting algorithm that calculates a fractional oil deficit metric to modulate choke adjustments more smoothly and avoid unnecessary oscillations. This approach defines a Dynamic Oil Target window – a variable range around the production setpoint – within which minor deviations are tolerated without triggering control actions.

If the actual oil production calculation remains within this fractional window, ROBOWELL® holds the choke position steady, promoting operational stability and reducing wear on actuators. This logic enables the system to distinguish between meaningful production deviations and normal process variability.

Mathematically, the fractional oil deficit is described as:

$$\text{Fractional Oil Deficit} = \text{fraction} * (\text{oil calculations} - \text{oil target})$$

This mechanism allows for more nuanced control, especially under fluctuating field conditions, and is a key enabler of autonomous, infrastructure-aware optimization. Figure 2, illustrates the concept of a variable tolerance band around the oil production setpoint. When production remains within this band, choke adjustments are suspended to maintain system stability.

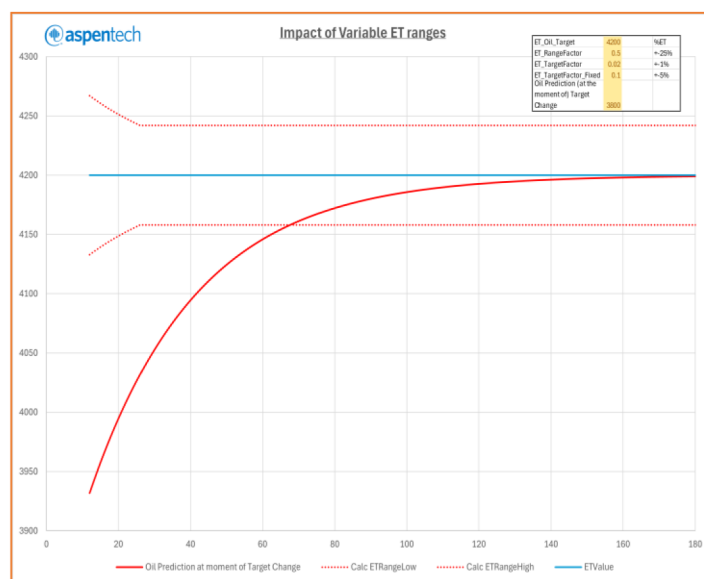


Figure 2—Dynamic Oil Target Window (Variable Target Range)

Together, these functions form an adaptive control strategy that integrates within a Model Predictive Control (MPC) framework to maintain operational safety and optimize production. The freezing logic acts as a hard override constraint to suspend choke movements during critical slug catcher surges, while the dynamic target window provides a flexible, data-driven approach to gradually modulate choke positions near production setpoints. This dual-layer control balances maximizing well output with safeguarding surface equipment integrity and process stability.

Appendix 1 presents the implementation of the model predictive control (MPC) logic. The controller dynamically adjusts choke positions based on real-time slug catcher levels and oil production targets. It demonstrates how autonomous decision-making is embedded at the edge, enabling safe and stable operation across variable flow conditions. The logic reflects the system's ability to freeze, hold, or adjust choke control in response to infrastructure constraints and production performance, forming a key part of the orchestration layer across the value chain.

Implementation Journey: From Concept to Field Reality

The transition from conceptual design to operational deployment of surface-aware control enhancements within ROBOWELL® required a structured, multi-phase approach. This journey encompassed stakeholder alignment, system integration, field validation, and iterative optimization, ultimately transforming theoretical control logic into a robust field-ready solution.

Conceptualization and Stakeholder Engagement

The initiative to integrate slug catcher level awareness into ROBOWELL®'s Advanced Process Control (APC) framework was driven by recurring surface facility disturbances linked to uncoordinated well choke movements. Early discussions with ADNOC operations highlighted the need for a control mechanism that could dynamically respond to surface constraints without compromising well productivity. These

conversations laid the foundation for defining operational thresholds and control objectives, ensuring alignment between subsurface optimization and surface integrity.

Simulation and Control Logic Validation

Before field deployment, the proposed freezing logic and Dynamic Oil Target Window algorithms were rigorously tested in simulation environments. Historical production data and synthetic slugging scenarios were used to evaluate system behavior under varying flow conditions. The freezing logic was validated to ensure it reliably halted choke movements when slug catcher levels exceeded critical thresholds, while the dynamic oil targeting algorithm was tuned to minimize unnecessary choke fluctuations near setpoints.

System Integration and Testing

Integrating the new control modules into the existing ROBOWELL® infrastructure required careful modification of the MPC-based choke optimizer. The freezing logic was embedded as a hard constraint, overriding all other optimization objectives when triggered. Simultaneously, the dynamic oil target window was implemented as a soft modulation layer, allowing smoother transitions in choke adjustments. Compatibility with legacy control systems was ensured through modular deployment, version control, and rollback protocols.

Field Deployment and Commissioning

Deployment followed a phased strategy, beginning with pilot wells selected based on historical slugging behavior and surface sensitivity. Real-time monitoring dashboards were developed to visualize slug catcher levels, choke positions, and control states, providing operators with transparency and confidence in the system's decisions. During commissioning, the system demonstrated its ability to prevent slug catcher surges by preemptively freezing choke movements, validating its effectiveness in live operations.

Challenges Faced During Deployment

Several challenges emerged during the deployment phase:

- **Threshold Calibration:** Determining appropriate slug catcher level thresholds required iterative tuning and close coordination with field teams to balance safety and operational flexibility.
- **Legacy System Compatibility:** Retrofitting the control logic into existing infrastructure demanded careful integration planning to avoid disruptions.
- **Sensor Reliability:** Real-time data accuracy was critical; sensor drift and latency issues necessitated the implementation of validation filters and fallback logic.
- **Operator Trust and Adoption:** Initial skepticism from field operators was addressed through training, visualization tools, and manual override options to build trust in automated control decisions.
- **Algorithm Sensitivity Tuning:** The fractional oil deficit algorithm required fine-tuning to avoid premature stabilization and ensure responsiveness without underproduction.
- **Multi-Well Coordination:** Synchronizing freeze logic across multiple wells feeding into shared facilities introduced complexity, prompting enhancements to support group-level control decisions.

Performance Outcomes and Learnings

Post-deployment analysis revealed significant improvements in surface facility stability, with a marked reduction in slug catcher level excursions and smoother choke transitions. Operator feedback was positive, citing increased confidence in system autonomy and reduced manual intervention. The implementation also highlighted the importance of integrating surface constraints into subsurface optimization, paving the way for more holistic field automation strategies.

Future Enhancements and Scalability

Building on the success of this deployment, future enhancements will focus on expanding surface-aware control to include additional constraints such as separator pressure and flare limits. The architecture also supports scalability across multi-well networks, enabling coordinated optimization that respects both subsurface and surface dynamics. This evolution positions ROBOWELL® as a foundational platform for autonomous field control in complex production environments.

Results: Smarter Decisions, Fewer Upsets

The implementation of slug catcher level freezing logic combined with a dynamic oil target window within the ROBOWELL® MPC framework has led to significant operational improvements. This integrated control strategy enables smarter, context-aware choke valve adjustments by incorporating real-time surface facility constraints, ensuring decisions are coordinated across wells. The freezing logic effectively prevents uncoordinated choke movements during slug catcher surges, while the dynamic target window minimizes unnecessary incremental adjustments near production setpoints. By adopting the dynamic target strategy during major oil target changes by Operations or FD teams, ROBOWELL is ensuring smooth transitions and avoiding sudden choke movements or downstream disturbances. These measures have reduced operational upsets, enhanced system stability, and decreased the frequency of manual interventions. Consequently, well and facility performance have improved, achieving optimized production within safe operational limits and fostering a more resilient, efficient offshore production environment.

The Choke Valve Movements – Shows the 50% reduction in daily choke valve movements. A 50% reduction in valve activity, thanks to smoother production control using the MPC dynamic oil target window logic. This reduces equipment wear and improves system reliability.

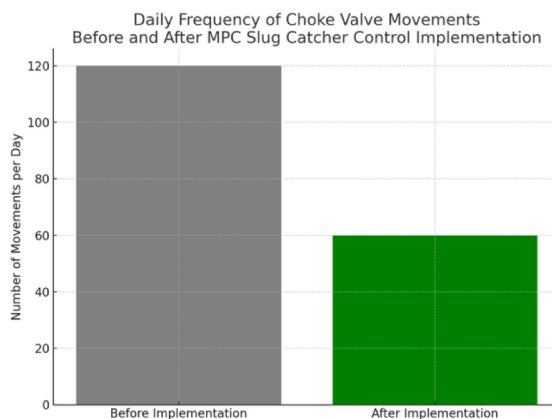


Figure 3—Choke Valve Movements before and after

Slug Catcher Excursions – Displays the decrease in both duration and peak level of slug catcher excursions. The level freeze logic reduced both the duration and peak level of slug catcher excursions, helping prevent carryover risks and protecting downstream processing systems.

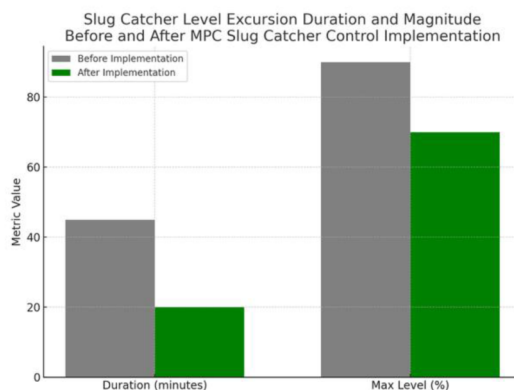


Figure 4—Slug Catcher Excursions

Reflections: What this Means for Automation

This project has redefined the boundaries of autonomous systems in upstream oil and gas operations. By bridging the traditionally silo-ed domains of subsurface intelligence and surface infrastructure, we have developed a system that operates with holistic awareness. The automation framework deployed around the slug catcher is not merely a control enhancement, it is a strategic enabler that transforms reactive flow management into proactive value chain orchestration.

The significance lies in the system's ability to align every component of the production chain, from wellhead to separator, with a unified objective: delivering stable, sustainable output within safe operating envelopes. This approach moves beyond incremental optimization and toward systemic coordination, where each asset contributes to the performance of the whole.

Central to this evolution is ROBOWELL®. Originally conceived as a well optimization tool, ROBOWELL® now functions as a value chain orchestrator. It interprets dynamic flow conditions, anticipates production instabilities, and coordinates responses across distributed assets, including slug catchers, robotic controllers, and surface facilities. This transition from localized automation to networked intelligence marks a critical inflection point in the digital transformation of upstream operations.

The implications for the oil industry are substantial. Automation is no longer limited to discrete tasks or isolated equipment. It is becoming a strategic layer embedded across operational domains, capable of autonomous calibration, adaptive control, and continuous learning. The architecture developed, low-infrastructure AI integrated with OT-native network models, demonstrates that scalable autonomy is achievable even in constrained environments.

Moreover, this project validates a broader vision: that upstream operations can be governed by intelligent systems capable of self-optimization, self-coordination, and self-protection. The autonomous slug catcher is not an endpoint, it is a proof-of-concept for a future in which production systems operate as intelligent ecosystems. As we look ahead, the same principles applied here can be extended to other critical scenarios, including gas lift optimization, flow assurance, and debottlenecking. Each deployment strengthens the foundation for a resilient, adaptive, and intelligent upstream infrastructure, one that is not only automated but autonomously governed.

Future Vision: Harmony Across the Chain

Imagine a future where every element of the oilfield—wells, facilities, pipelines, compressors, and beyond; operates not in isolation, but in harmony. A future where decisions are not made in silos, but in sync. This project demonstrates that such a future is not only possible, it is within reach. By embedding intelligence across the production landscape, we have shown that upstream and downstream systems can be unified

through shared logic, real-time data, and a collaborative mindset. The result is a production environment that is not only more efficient, but fundamentally more resilient and adaptive.

This initiative did more than solve a technical challenge. It redefined how we think about automation. We moved from optimizing individual components to orchestrating entire systems. From reactive control to predictive coordination. From fragmented operations to integrated performance.

The vision ahead is clear: a fully connected value chain where autonomous agents continuously align production goals with operational constraints, environmental considerations, and market dynamics. In this future, automation is not a layer added to operations, it is the foundation upon which operations are built. This is the next frontier for the energy sector. And this project is a step toward that horizon.

References

1. Hernandez, C., N. Matta, W. Almadhoun, et al 2024. "Revolutionizing Oil Field Operations: Implementing Multivariable Predictive Control to Drive Autonomous Well Control Operations." ADIPEC, November 4, D031S109R003. <https://doi.org/10.2118/222967-MS>.

Appendix 1

Sample of the logic implemented

```
python

class RoboWellMPCController:
    def __init__(self, slug_catcher_threshold, oil_setpoint,
target_window_fraction):
        self.slug_catcher_threshold = slug_catcher_threshold
        self.oil_setpoint = oil_setpoint
        self.target_window_fraction = target_window_fraction
        self.choke_control_frozen = False
        self.current_choke_position = 0.0

    def monitor_slug_catcher_level(self, level):
        if level > self.slug_catcher_threshold:
            self.choke_control_frozen = True
        elif self.choke_control_frozen and level <= self.slug_catcher_threshold:
            self.choke_control_frozen = False
        return self.choke_control_frozen

    def calculate_fractional_oil_deficit(self, actual_production):
        return (self.oil_setpoint - actual_production) / self.oil_setpoint

    def apply_dynamic_oil_target_window(self, fractional_deficit):
        lower_bound = -self.target_window_fraction
        upper_bound = self.target_window_fraction
        return lower_bound <= fractional_deficit <= upper_bound

    def mpc_optimize_choke(self, slug_catcher_level, actual_oil_production):
        if self.monitor_slug_catcher_level(slug_catcher_level):
            # Freeze choke movements when slug catcher level is high
            return self.current_choke_position, "Freeze choke control due to high
slug catcher level"
        fractional_deficit =
self.calculate_fractional_oil_deficit(actual_oil_production)
        if self.apply_dynamic_oil_target_window(fractional_deficit):
            # Hold choke position within the dynamic target window
            return self.current_choke_position, "Hold choke - production within
target window"
        # Adjust choke position incrementally based on deficit
        adjustment = fractional_deficit * 10.0
        self.current_choke_position += adjustment
        return self.current_choke_position, "Adjust choke position"

# Example usage
controller = RoboWellMPCController(75.0, 1000.0, 0.05)
choke_pos, status = controller.mpc_optimize_choke(80.0, 950.0)
print(f"Choke Position: {choke_pos}, Status: {status}")
```